21 November 2018 (Wednesday 9am)

Experiment 6 & 7

Harmonic Oscillator Part II: Physical Pendulum

Waves on a Vibrating String

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Abstract

Analysis of the Harmonic Motion of a Physical Pendulum and a Vibrating String

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Rigid objects and non-rigid, spatially expansive systems exhibit harmonic motion, in which a sinusoidal function describes the motion of the system. The experiment seeks to analyze the motion of a physical pendulum within a frequency domain and analyze the three regimes of damping that an oscillating system may experience: underdamped motion, critically-damped motion, overdamped motion. In addition, the experiment seeks to quantify the resonance $\omega_0 = (0.7463 \pm 0.0223)Hz$, damping time, and quality factor of the system. Similarly, the experiment studies the harmonic oscillations of a non-rigid, spatially extensive systems such as the waves on a vibrating string. To analyze the harmonic motion of a vibrating string, the system will be observed while under the experience of the fundamental resonant frequency $f_0 = (7.4710 \pm 0.0005)Hz$, varying tensions, varying frequencies, and varying boundary conditions.

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Introduction

Harmonic motion occurs due to the periodic motion of an object in a sinusoidal pattern influenced by damping and driving forces. Jean le Rond d'Alembert proposed the first equation representing the solution to the motion of a vibrating string. Later on, his student Pierre-Simon Laplace would go on to pioneer the Laplace transform that takes a function of time to a function of frequency. Two experiments will be conducted to observe and analyze the harmonic motion of a rigid object such as a physical pendulum and a non-rigid, spatially expansive system such as a vibrating string. The first experiment will focus on studying the harmonic motion of a physical pendulum on a frequency domain. By applying different damping force magnitudes on a physical pendulum undergoing harmonic motion, the damped harmonic motion can be categorized into three regimes of damping using the undamped resonance frequency of the oscillations (ω_0):

a) Underdamped motion: $\omega_0 < \frac{1}{\tau}$. ω_{damped} is real due to the presence of oscillations.

b) Overdamped motion: $\omega_0 > \frac{1}{\tau}$. ω_{damped} is imaginary due to the absence of oscillations.

c) Critically-damped motion: $\omega_0 = \frac{1}{\tau}$. ω_{damped} is imaginary similar to b.

Driven, damped oscillations of a physical pendulum can be similarly studied using parametric plots called Lissajous figures that map the amplitude and the angular displacement of the oscillations. With the aid of a wave driver, the oscillations of the physical pendulum can be driven at certain frequencies. The oscillations at these frequencies will produce certain Lissajous figures that can aid in the determination of the resonant frequency. Additionally, the amplitudes of the oscillations at these frequencies will produce a resonance curve that relates the oscillation amplitude to the oscillation frequency.

The second experiment focuses on the vibrational motion produced by a massive string. The experiment utilizes the actuator of a wave driver to periodically produce a force in the vertical direction, which will induce oscillations. To measure the vertical displacement of the string, a laser light is pointed to the string and scattered into a photodiode. By applying different tensions on a stretched string and measuring the corresponding linear densities of the stretched string, the wave speed on the string can be calculated. Conceptually, the wave speed of the stretched string, corresponding to the respective tension, can be studied using a graph that plots the periodic movement of a pulse moving along the stretched string. By monitoring the maximum amplitude of the vibrating string using a Lissajous figure, the resonant frequency can be determined and studied by changing the boundary conditions of the driven oscillations.

Methods

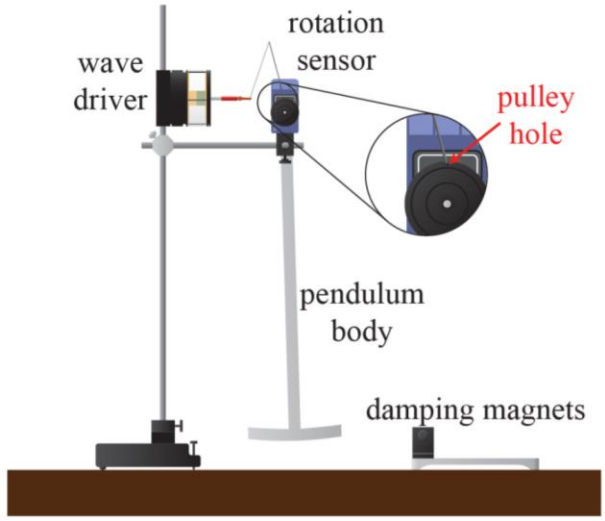


Figure 1.1. Experiment Setup to investigate a Physical Pendulum's damping regimes and its resonance. The damping magnets serve to produce a tunable velocity-dependent damping force. The wave driver coupled to the pendulum by a torsion spring serves to produce a tunable frequency-dependent driving force.

The Physical Pendulum Experiment analyzes the harmonic motion of a rigid object under the effect of a damping force and then the combined effects of damping and driving forces. As shown in Figure 1.1, the experiment will use a physical pendulum, damping magnets, a rotation sensor, and wave driver connected to a torsion spring. The first part of this experiment uses a software configuration that monitors the time and the angular displacement of the oscillator at a sampling rate of 50 Hz on continuous mode. At the start of each trial, the software must be configured to record angular readings at the equilibrium position of the pendulum. Optionally, the starting condition of the oscillation can be recorded using a photogate. To analyze the effects of damping on the oscillations, the gap between the magnets will be altered for five trials (10mm, 11mm, 15mm, 30mm, 35mm) to study the three regimes of damping: overdamped, critically-damped, and underdamped. For each gap, the pendulum would be pulled from a starting position and let go such that the pendulum's angular displacement does not exceed the width of the curved bottom of the pendulum. The pendulum is displaced such that frequency of the oscillations will be approximately independent of amplitude for "small" amplitudes.

The second part of the first experiment examines the amplitude and the phase response of a driven, damped oscillator. With the data from the last part of the experiment, the gap between the magnets must be altered such that the damping force causes the oscillations to damp out in 10 seconds if undriven (28mm). By attaching banana plugs from the wave driver into the output of the DAQ measurement tool (850), a Signal Generator function is used to produce a sinusoidal driving torque on the pendulum at 5 V. By plotting Lissajous figures, the resonance frequency will be determined. Additionally, a resonance curve is plotted by measuring the amplitude responses of the oscillator corresponding to certain frequencies (0.400-1.032 Hz).

Figure 1.2. Experiment Setup to investigate the harmonic motion of a stretched vibrating string. The setup uses a wave driver to induce periodic vertical oscillations. A photodiode along with a laser beam monitors the intensity of the scattered light illuminated on the string. A mass is hung from a pulley to stretch the string.

The Vibrating String Experiment examines the properties of the vibrational motion of a non-rigid, spatially expansive system under the effects of varying tensions and varying boundary conditions. The lab instruments used in this experiment are indicated in figure 1.2. Two clamps will be placed at two ends where a wave driver is located at one end and a pulley-mass system is located at the other end. The wave driver will excite vibrations by moving an actuator and periodically applying a force vertically. To track the vertical displacement of the string, a laser beam, placed 8 cm away from the pulley apex, illuminates the top third of the string and a photodiode, placed above the laser spot with a gain setting of 10, tracks the intensity of the light scatter.

The first part of the experiment predicts and measures the wave speed for varying tensions. To predict the value of the wave speed, the linear mass density of the stretched string must be calculated by measuring the mass of the stretched string M without the mass of the limping portion of the string. With the mass value, the linear mass density is calculated by dividing M by the total stretched length of the string for three tension values (2.45N, 2.94N, 3.92N). To measure the wave speed using DAQ software, the configuration will record the time, output voltage, and light intensity using Continuous mode. A signal generator function, set to Auto, will be used generate Square waves at a 0.2 Hz frequency and at a 2 V amplitude. The data will be recorded for the same three tension values used in the prediction of the wave speed value.

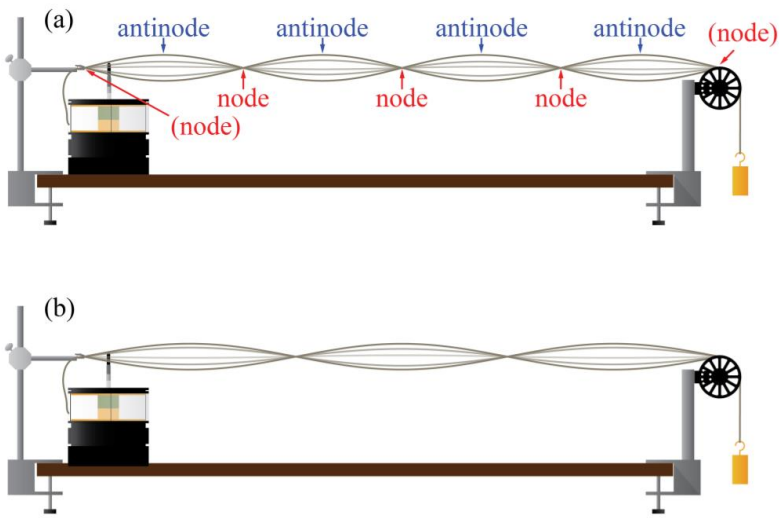


Figure 1.3. Visual of the harmonic motion of a vibrating string. Nodes and Antinodes will be observed during the string vibrations. They will increase as the mode of the frequencies increase.

The second part of the experiment examines the harmonic motion of the string vibrating at resonant frequency and frequencies of higher modes (n = 1, 2, 3, 4, 5, 6). The resonant frequency for the fundamental mode (7.473 Hz) is chosen by plotting a Lissajous figure of light intensity and output voltage. The third part of the experiment examines the same harmonic motion at frequencies of higher modes (n = 2, 4, 5), but with varying boundary conditions. In both of these experiments, the amplitude of the oscillations will be analyzed by measuring the intensity of the light scattered on the string. For the third part of the experiment, the vibrating string will be constrained in the middle to add a node in the middle of the vibrating string.

Analysis

Harmonic Motion of a Physical Pendulum (Damped)

As shown in figure 1.1, a physical pendulum is released at some height and experiences various torques that causes the rigid object to oscillate. Given the angular displacement of the pendulum from its equilibrium configuration θ, the mass of the pendulum M, and the location of its center of mass below the rotation axis, the solution for the motion of the pendulum can be derived. The components to the solution are the torques due to gravity (τ_g), an attached spring (τ_{spr}), and the magnets ($\tau_{damping}$). Since the physical pendulum will only experience small angular displacements, small angle approximations will be invoked to approximate the trigonometric terms in the solution.

$$I\ddot{\theta} = \sum_i \tau$$

$$\tau_{net} = \left(-\tau_g\right) + \left(-\tau_{spr}\right) + \left(-\tau_{damping}\right) = -\left(\tau_g + \tau_{spr} + \tau_{damping}\right)$$

$$\tau_{net} = -\left(MgLsin(\theta) + Bsin(\theta)cos(\theta) + b\dot{\theta}\right)$$

For $(\theta \ll 1) \Rightarrow sin(\theta) \approx \theta, cos(\theta) \approx 1$

$$\tau_{net} \approx -\left(MgL\theta + B\theta + b\dot{\theta}\right)$$

$$\tau_{net} = -\theta(MgL + B) - b\dot{\theta}$$

$$\tau_g = F \times r = Frsin(\theta) = MgLsin(\theta)$$

$$\tau_{spr} = Bsin(\theta)cos(\theta)$$

$$\tau_{damping} = b\dot{\theta}$$

$$k \equiv MgL + B$$

$$\tau_{net} \approx -\theta k - b\dot{\theta}$$

$$I\ddot{\theta} = -\theta k - b\dot{\theta}$$

This equation of motion will be converted into complex notation $\left(\theta(t) = Ae^{jwt}\right)$ by comparing the solution to the motion equation of a linear damped oscillator and substituting the following values: $x \Rightarrow \theta, m \Rightarrow I$. Furthermore, the damped and undamped resonance frequencies $\left(\omega_{damped}, \omega_0\right)$ and the damping time (τ) can be determined based on the same components derived using the motion equation of a linear damped oscillator.

Derivations for angular position, velocity, and acceleration:

$$\theta(t) = Ae^{j\omega t}$$
$$\dot{\theta}(t) = i\omega Ae^{i\omega t}$$
$$\ddot{\theta}(t) = -\omega^2 Ae^{i\omega t}$$

Complex solution for the oscillator:

$$x(t) = Ae^{jwt}$$
$$x \Rightarrow \theta$$
$$\theta(t) = Ae^{jwt}$$

Solution for ω:

$$-\omega^2 mAe^{jwt} = -kAe^{j\omega t} - j\omega Ae^{j\omega t}$$
$$m\omega^2 = k + j\omega b$$
$$\omega = \frac{jb}{2m} \pm \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$$
$$m \Rightarrow I$$
$$\omega = \frac{jb}{2I} \pm \sqrt{\frac{k}{I} - \frac{b^2}{4I^2}}$$

Imaginary and real solution of an undriven, damped oscillator:

$$x(t) = Ae^{j\sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}t} \times e^{-bt/2m}$$
$$x \Rightarrow \theta, m \Rightarrow I$$
$$\theta(t) = Ae^{j\sqrt{\frac{k}{I} - \frac{b^2}{4I^2}}t} \times e^{\frac{-bt}{2I}}$$

Solution for the undamped resonance frequencies $(\omega_{damped}, \omega_0)$ and the damping time (τ):

$$\omega_{damped} = \sqrt{\frac{k}{I} - \frac{b^2}{4I^2}} = \sqrt{\omega_0^2 - \frac{1}{\tau^2}}$$
$$\omega_0 \equiv \sqrt{\frac{\mathrm{k}}{\mathrm{I}}}$$
$$\tau \equiv \frac{2\mathrm{I}}{\mathrm{b}}$$

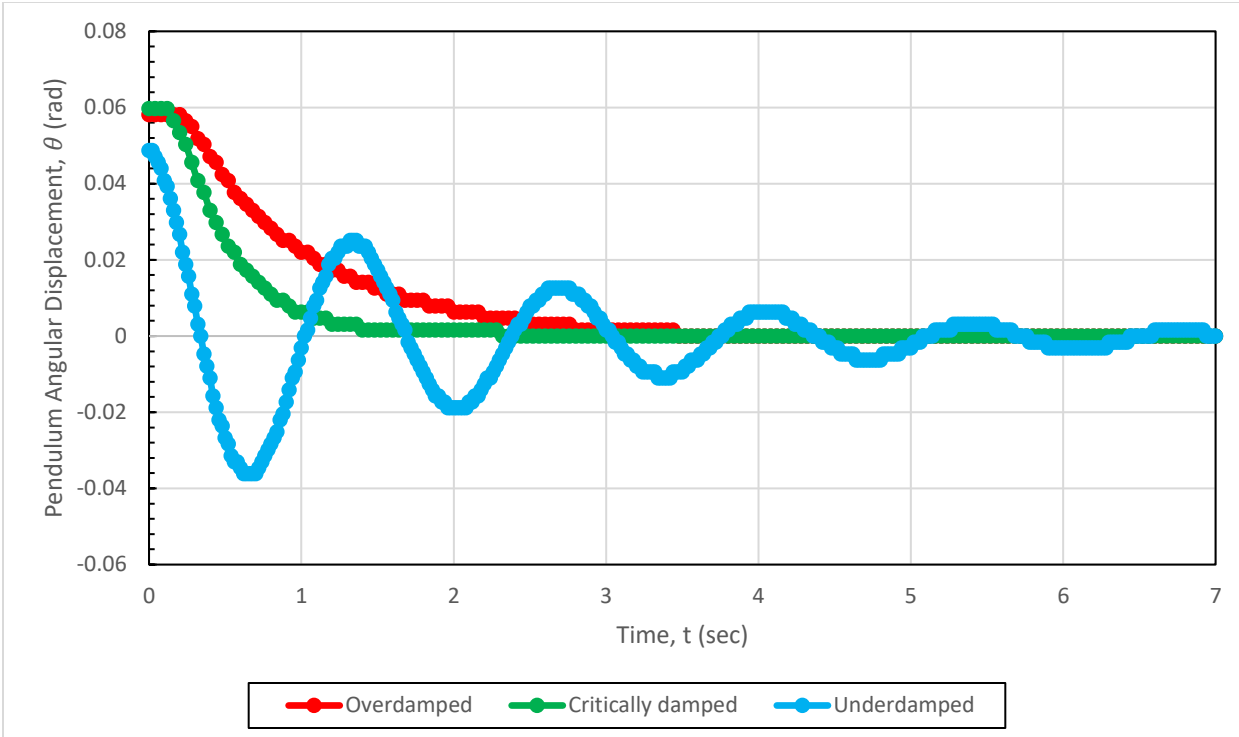


Figure 1.4. Pendulum angular displacement over time of an undriven, damped pendulum based on the regimes of damping. The damped harmonic motion of a physical pendulum was plotted based on the damping forces due to two magnets with gaps of the following distances $d = [(10.0,15.0,30.0) \pm (0.5)]mm$.

Regime	Distance between magnets (cm)
Underdamped	1.00 ± 0.05
Overdamped	1.50 ± 0.05
Critically-damped	3.00 ± 0.05

Table 1.1. Damping regimes based on the distances between the two magnets.

Using the experiment setup indicated in figure 1.1, the harmonic motion of a physical pendulum while under the effect of a damping force due to the eddy currents produced when the physical pendulum, acting as a conductor, moves through a magnetic field produced by two magnets. The damping force was varied for five different gap distances between the two magnets to distinguish the damping regimes experienced by the physical pendulum for each distance. The five distances used were $d = [(10.0,11.0,15.0,30.0,35.0) \pm (0.5)]mm$. The criteria used for relating the gap distances d to the damping regimes based on wave functions is as follows:

a) Underdamped $\left(\omega_0 < \frac{1}{\tau}\right)$: slowest exponential decay with oscillations.

b) Overdamped $\left(\omega_0 > \frac{1}{\tau}\right)$: median exponential decay without oscillations.

c) Critically damped $\left(\omega_0 = \frac{1}{\tau}\right)$: fastest exponential decay without oscillations.

Observations were made for the undamped, undriven oscillator to determine the undamped oscillation frequency ω_0. To do so, the inverse of the difference of the time positions between the first two maxima, observed in the figure 1.4, was calculated as the best value while the uncertainty was calculated using the uncertainty equation for products and ratios where $\delta T = 0.04$.

$$\delta\omega_0 = \omega_{0,best}\left(\frac{\delta T}{T_{best}}\right)$$

$$T^{-1} = (2.72 - 1.38)^{-1} \Rightarrow \omega_0 = (0.7463 \pm 0.0223)Hz$$

Harmonic Motion of a Physical Pendulum (Driven, Damped)

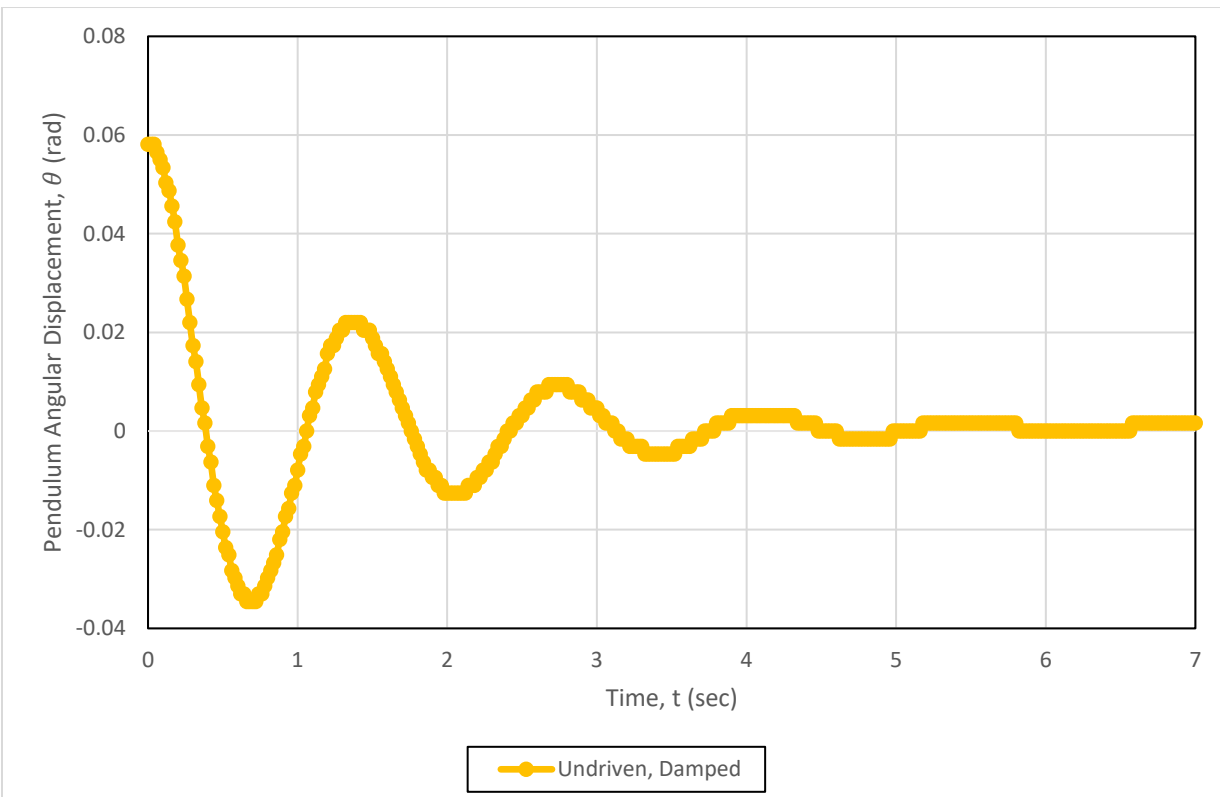


Figure 1.5. Time trace of an undriven, damped physical pendulum. An undriven, damped physical pendulum was observed with a magnet gap distance of $d = (28.0 \pm 0.5)mm$.

The mean damping time τ_{avg} is calculated using T and the ratio of successive amplitude measurements. The uncertainty was calculated using the procedure outlined in §ii.1.6.

$$\tau_{avg} = -\frac{T}{ln\left[\frac{A(t+T)}{A(t)}\right]} = (1.440 \pm 0.0036)s$$

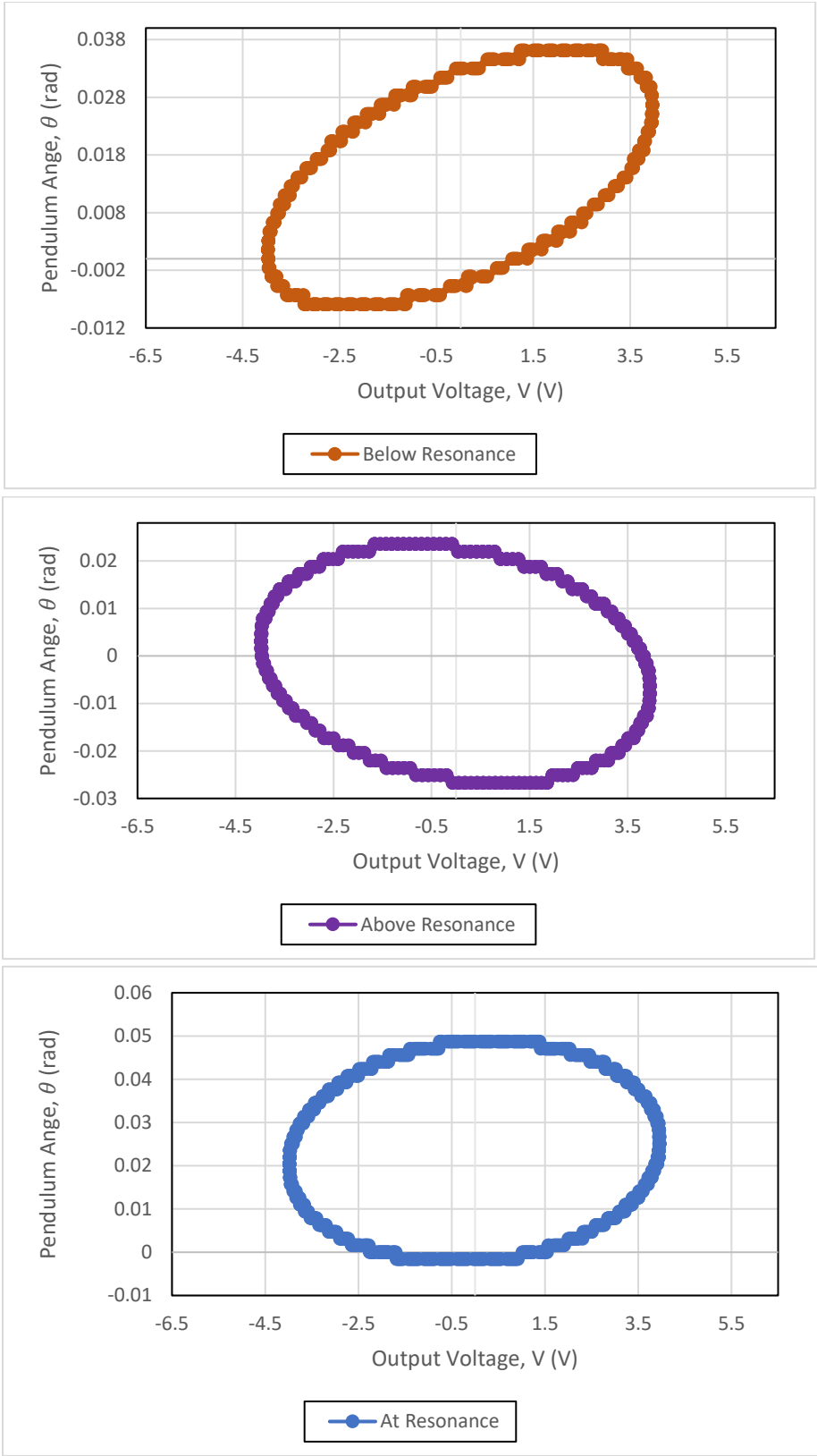


Figure 1.6. Lissajous figures of the driven, damped physical pendulum. The pendulum was driven at an amplitude of 5 V and [0.650 (below), 0.716 (at), 0.750 (above)] Hz.

The resonant frequency of a damped, driven system ω_R can be determined by differentiating the amplitude of the particular solution to the motion of the system with respect to ω_d.

$$\theta_p(t) \equiv \frac{cos\big(\omega_d t - \varphi(\omega_d)\big)}{\sqrt{(\omega_0^2 - \omega_d^2)^2 + \left(\frac{2\omega_d}{\tau}\right)^2}}$$

Definition of the resonant frequency of a damped and driven harmonic oscillator (ω_R):

$$\omega_R = \sqrt{\omega_0^2 - 2\frac{1}{\tau^2}}$$

In addition to the definition of ω_R, the Lissajous plots shown in figure 1.6 indicates the resonant frequency of the damped, driven physical pendulum through a parametric plot of the output voltage and the angular displacement of the physical pendulum. At resonance, the Lissajous figure represents the harmonic motion of a damped, driven oscillator as a circle. By referring to the third plot of figure 1.6, the ω_R is approximately equal to (0.7160 ± 0.0005) Hz.

Definitions of the Quality Factor Q:

$$Q = \frac{\sqrt{km}}{b} = \frac{\sqrt{kI}}{b} \equiv \frac{1}{2}\tau\omega_R$$

$$Q \approx \frac{\omega_0}{\Delta\omega}$$

Using the calculated and measured values for τ and ω_R, the quality factor Q can be calculated using its definition. Through this method, the quality factor Q is approximately equal to (0.5200 ± 0.0014). Another method for the calculation of the quality factor Q is to calculated the ratio of the resonant frequency and the full width of the resonance where the amplitude is $1/\sqrt{2}$ of the maximum amplitude $(\Delta\omega)$. The points, and the resonance width, can be determined using the resonance curve in figure 1.7. The ratio, also Q, is calculated to be (2.7224 ± 0.0055). Based on the calculated values of Q from the two definitions of the quality factor, the values deviate each other by a magnitude of 2.204, which does not fall within either value's error-of-margin. This can be attributed to the approximations made in the derivations such as the small angle approximations and in the inaccurate choices for the frequencies chosen for the resonance width.

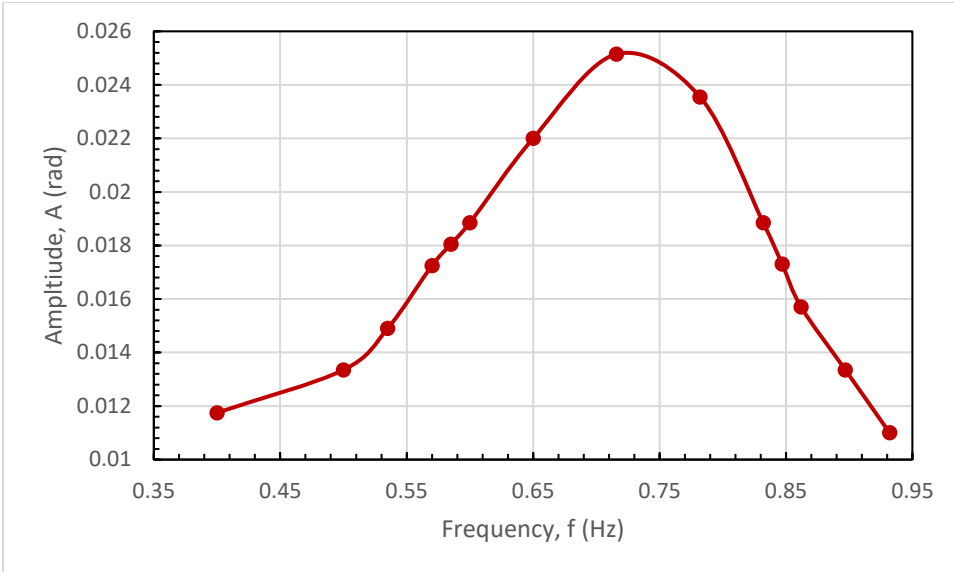


Figure 1.7. Oscillation Amplitude Spectrum of a damped, driven oscillator. By joining the amplitude points found at certain frequencies, a Lorentzian shape is formed.

Harmonic Motion of a Vibrating String

Given the experiment setup shown in figure 1.2, the tension-dependent wave speed can be predicted and measured. For this experiment, a string of mass 0.015 kg and length of 2.245 m with a linear mass density of 0.0067 kg/m is used. The prediction of the wave speed is given by the equation where T is the tension value and μ is the linear mass density of the string:

$$v = \sqrt{\frac{\mathrm{T}}{\mu}}$$

Hanging Mass (kg)	Tension (N)	Linear Mass Density (kg/m)	Wave velocity (m/s)
0.2500 ± 0.0005	2.450 ± 0.005	0.00650 ± 0.00001	19.4145 ± 0.0496
0.3000 + 0.0005	2.940 ± 0.005	0.00630 ± 0.00001	21.6025 ± 0.0503
0.4000 ± 0.0005	3.920 ± 0.005	0.00600 ± 0.00001	25.5604 ± 0.0536

Table 1.2. Calculations for wave speeds for each hanging mass. Uncertainty was calculated using the uncertainty formula for products and ratios where $\delta g = 0.0003 m/s^2$.

$$\delta v = |v_{best}| \sqrt{\left(\frac{\delta T}{T_{best}}\right)^2 + \left(\frac{\delta \mu}{\mu_{best}}\right)^2}$$

Another method of calculating the wave speed is by using the same setup in figure 1.2. For this calculation, the light intensity measured by a photodiode is compared to the time recorded by the DAQ software. The following equation will be used for the calculation where L is the length of the stretched string:

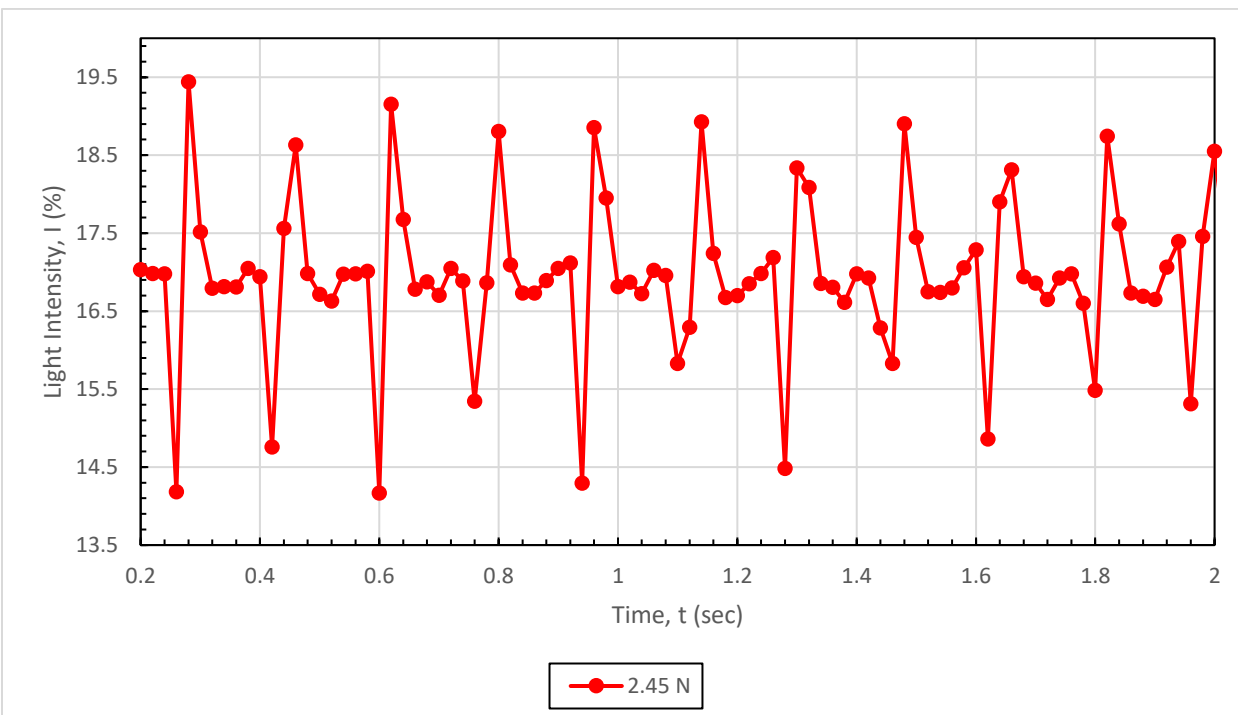
$$v = \frac{2L}{t}$$

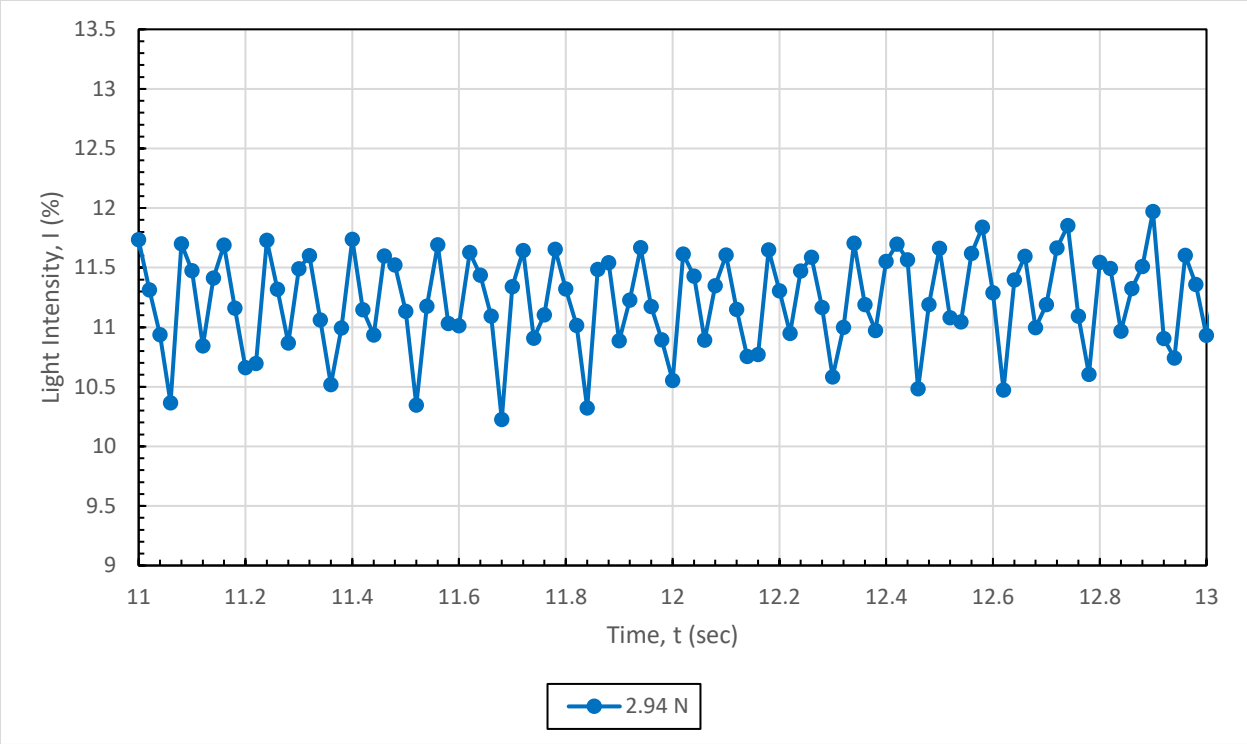
Hanging Mass (kg)	Twice of Stretched String Length (m)	Time interval sharp rising edges (s)	Wave velocity (m/s)
0.2500 ± 0.0005	3.4000 ± 0.0005	0.180 ± 0.005	18.8889 ± 0.5247
0.3000 + 0.0005	3.4000 ± 0.0005	0.160 ± 0.005	21.2500 ± 0.6641
0.4000 ± 0.0005	3.4000 ± 0.0005	0.140 ± 0.005	24.2857 ± 0.8673

Table 1.3. Calculations for the wave speed using the travel distance and time interval of a pulse. Uncertainty was calculated using the following equation:

$$\delta v = |v_{best}| \sqrt{\left(\frac{\delta m}{m_{best}}\right)^2 + \left(\frac{\delta t}{t_{best}}\right)^2}$$

Based on the results of the two methods of calculations for the wave speed, the data agrees with each other's errors-of-margin and suggests that the data is accurate.





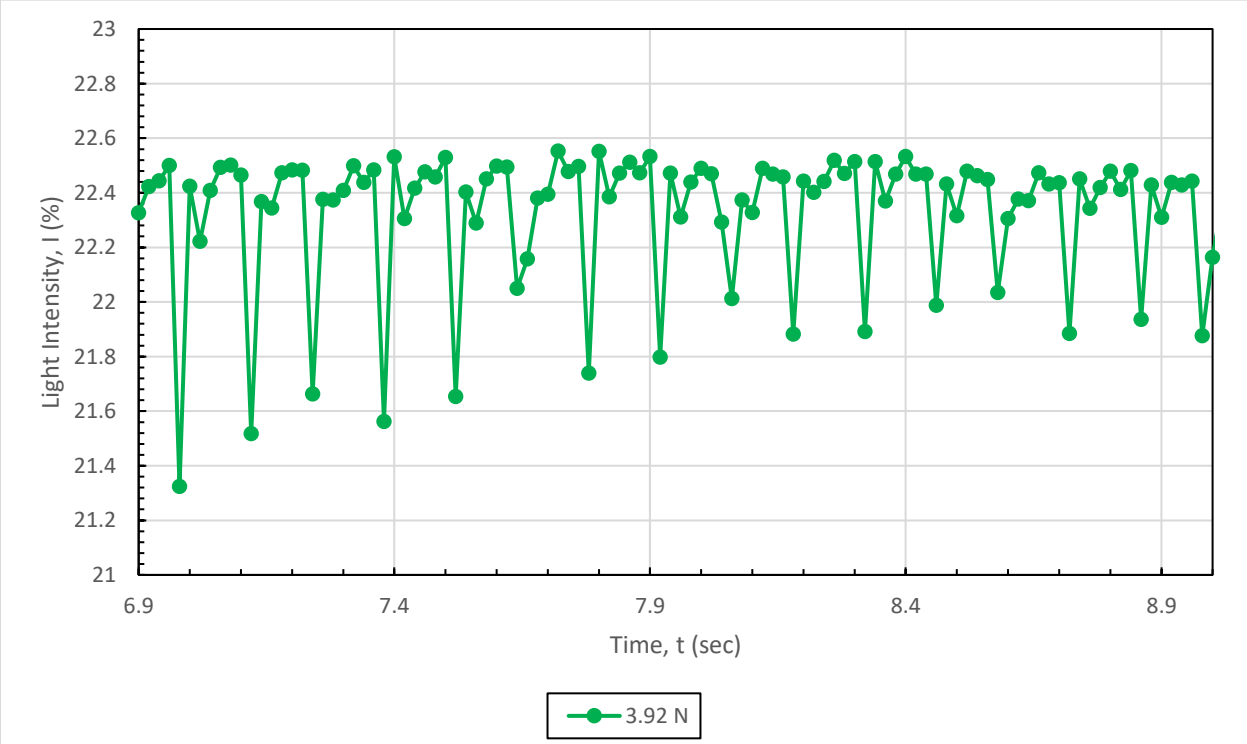


Figure 1.8. Plots of the periodic reflections of a pulse across the length of the stretched string. Time positions were taken to calculate for the wave speed. The stretched string was observed for the following hanging masses and their respective tension values: (0.25, 0.30, 0.40) kg, (2.45, 2.94, 3.92) N.

To determine the fundamental resonant frequency of a vibrating string, a software was configured to produce a Lissajous figure that compare the output voltage recorded by the wave driver and the light scatter intensity captured by the photodiode. Through trial-and-error, the shape and size of the Lissajous figure to approximate a circle. By doing so, the fundamental resonant frequency for the vibrating string is determined. Then, the string was driven at certain modes to compare calculated and measured values for the frequency at a certain mode.

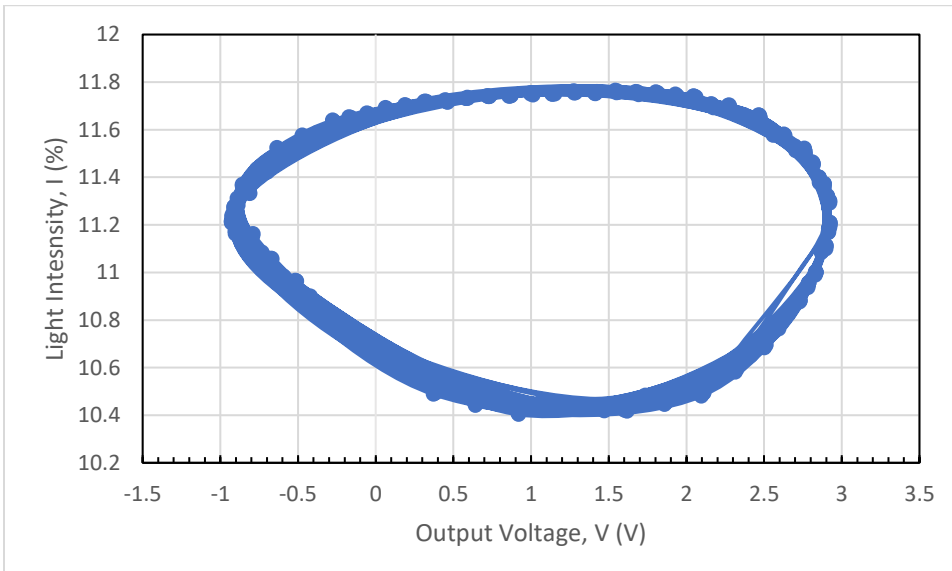


Figure 1.9. Lissajous Plot of the Fundamental Resonant Frequency. The vibrating string was driven at $(7.4710 \pm 0.0005)Hz$. The lack of tilting in the shape indicates that the driven frequency is the fundamental resonant frequency.

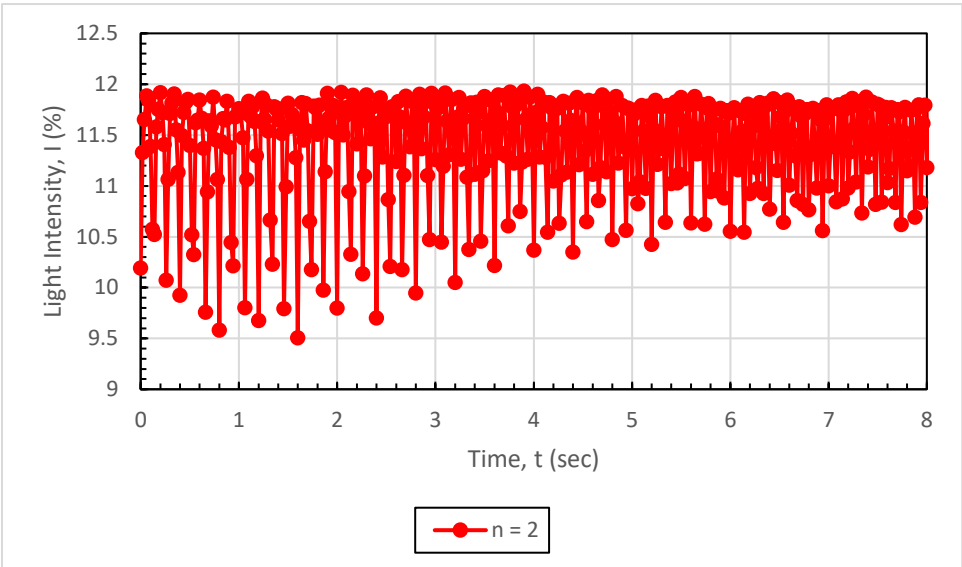
Mode (n)	Measured Frequency (Hz)	Calculated Frequency (Hz)
1	7.4710 ± 0.0005	7.1423 ± 0.2551
2	14.9460 ± 0.0005	14.2857 ± 0.0182
3	22.4190 ± 0.0005	21.4286 ± 0.0273
4	29.8920 ± 0.0005	28.5714 ± 0.0364
5	37.3650 ± 0.0005	35.7143 ± 0.0455
6	44.8380 ± 0.0005	42.8571 ± 0.0547

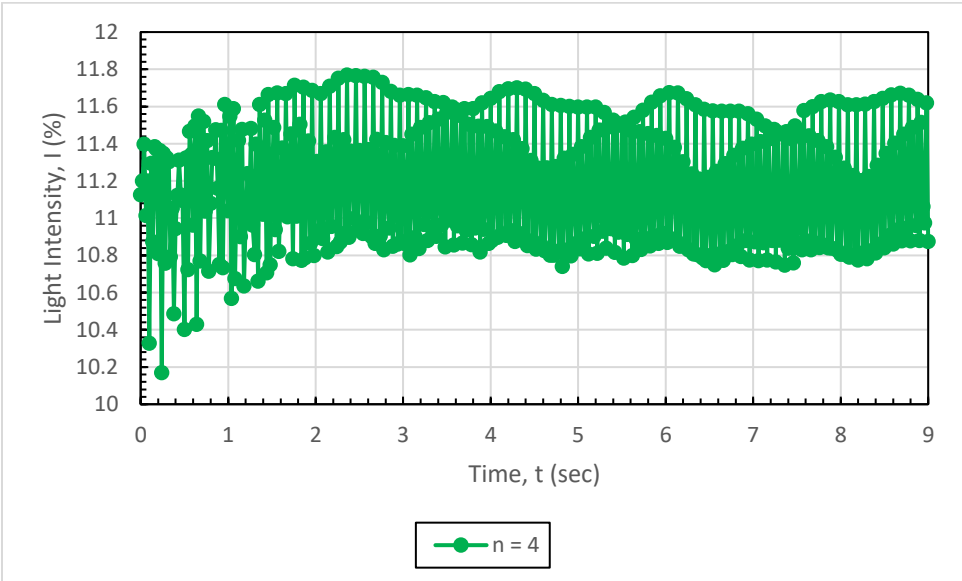
Table 1.4: Data of measured and calculated frequencies for n^{th} modes using the equation:

$$f(n) = \frac{nv}{2L}.$$

Based on the results of table 1.4, the values for the calculated frequencies were less than the measured frequencies. Furthermore, none of the measured frequency values fell in the margin of error for the calculated frequency values. However, the deviation for each matching value is less than 4.5%. Therefore, the data set is still valid.

The final portion of the experiment studies the effect of placing a node in the middle of a vibrating string for the modes $n = 2,4,5$. For modes 2 and 4, a node is located in the middle of the vibrating string. Therefore, no interference is expected to occur for these modes. However, for mode 5, an antinode is located in the middle. Hence, interference is expected to influence the amplitude of the wave.





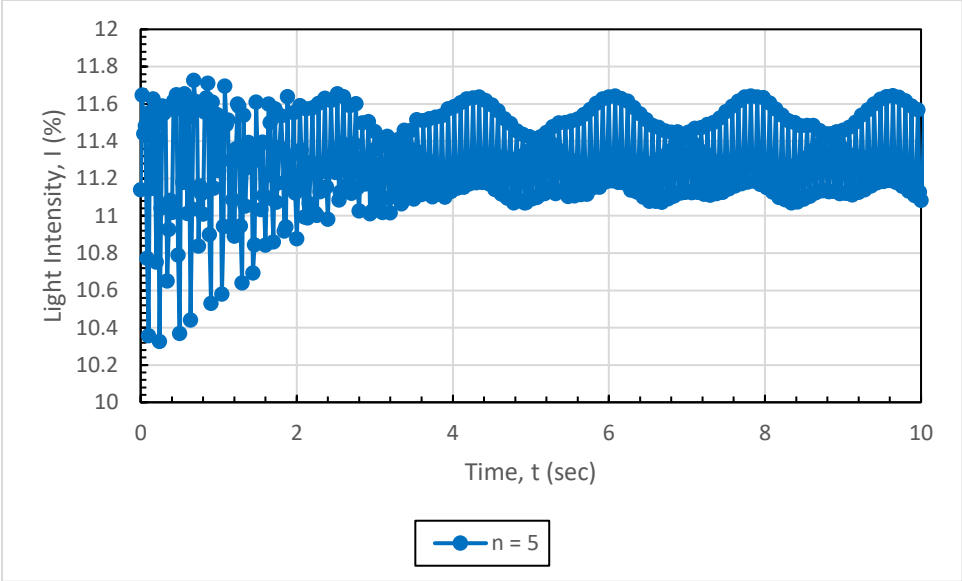
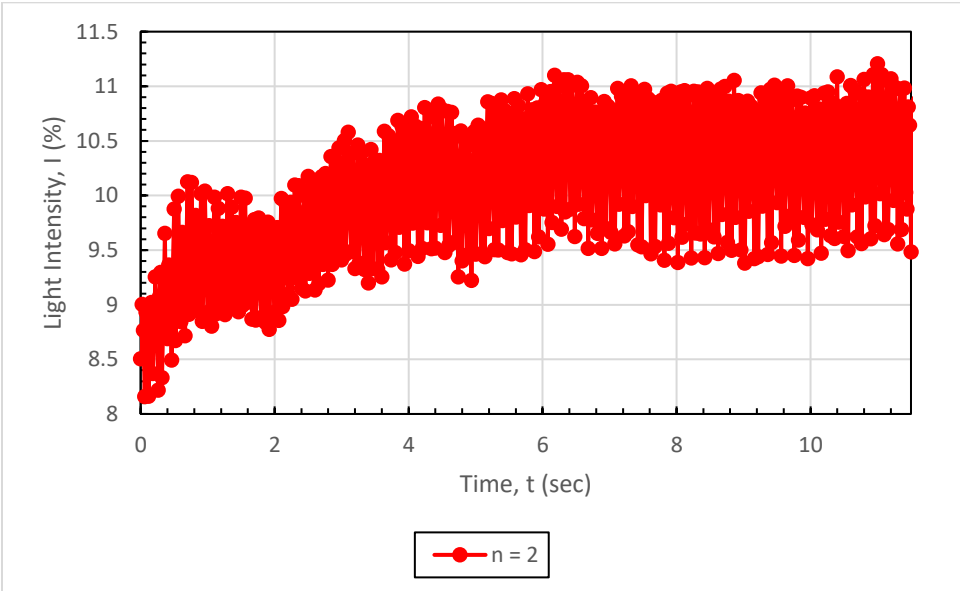


Figure 1.10. Amplitude of the vibrating string over time at modes n = 2, 4, and 5 without adding a node in the middle.



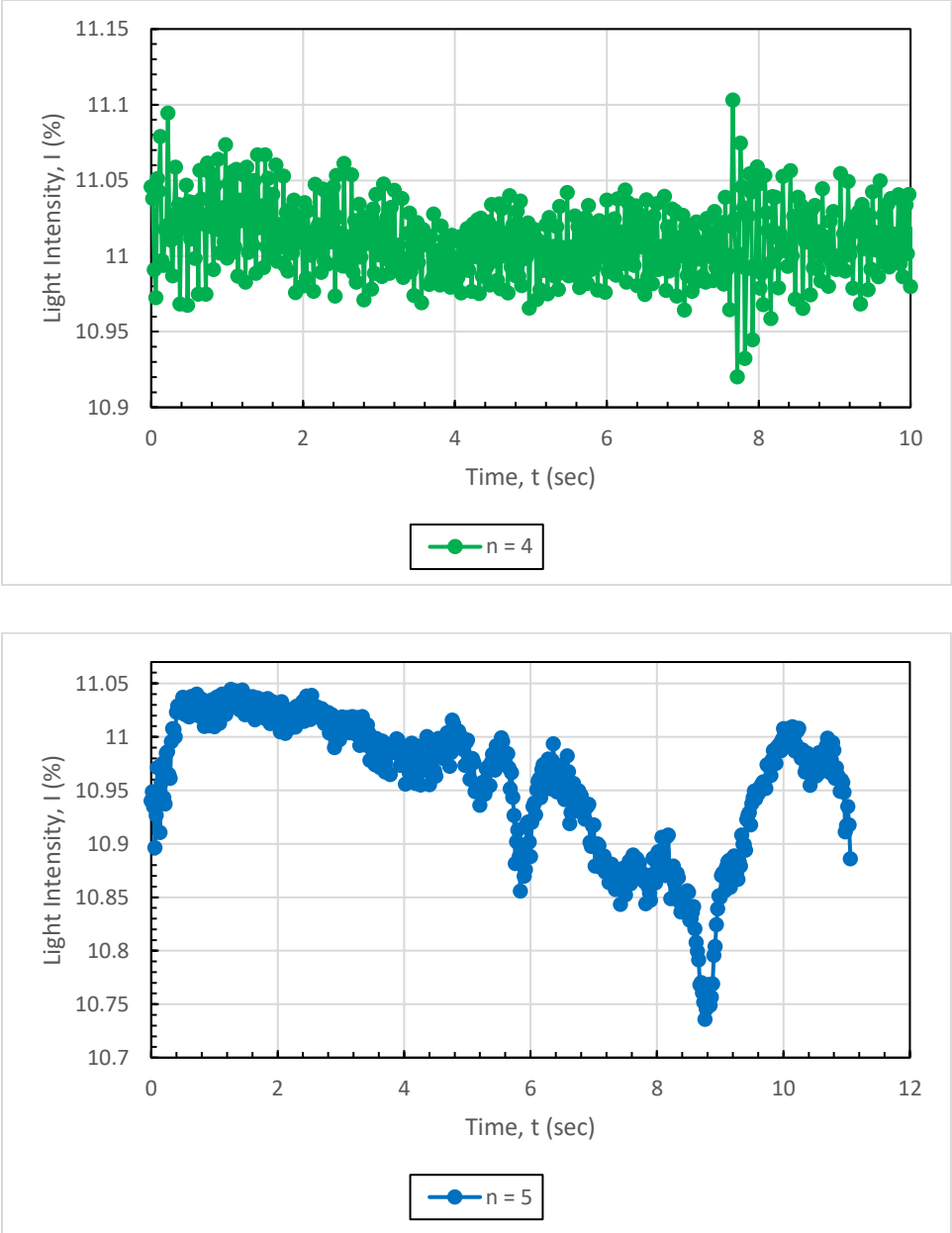


Figure 1.11. Amplitude of the vibrating string over time at modes n = 2, 4, and 5 with an added node in the middle.

Based on figures 1.10 and 1.11, a comparison can be made for the amplitude changes resulting from an added middle node. For modes 2 and 4, the shape of each graph closely matches the shape of their corresponding graphs. However, the amplitude suddenly spiked in mode 4. This spike is attributed due to the lack of stability of a human hand holding the middle of a vibrating string. For mode 5, the drastic reduction in amplitude was expected due to the interference caused by adding a middle node. For the last graph in figure 1.11, the amplitude for mode 5 dropped by 2.23 %.

Conclusions

The experiment's goal was to study the harmonic motion of a rigid object such as a physical pendulum and the vibrational motion of a non-rigid, spatially expansive system such as a string. The first part of the experiment focuses on the three regimes of damped motion: underdamped, critically damped, and overdamped. The distance between two magnets were varied to exert different damping forces on the oscillating pendulum and the movement of the physical pendulum were categorized accordingly. The second part of the experiment focuses on the resonant frequency of a vibrating string. Then, the behavior of the string was examined for different modes and different boundary conditions.

Based on five trials, three gap distances were classified as underdamped $(1.00 \pm 0.05)cm$, critically damped $(1.50 \pm 0.05)cm$, and overdamped $(3.00 \pm 0.05)cm$. With the graphs plotted based on the motion of the pendulum under different regimes, a period was determined for the calculation for the undamped oscillation frequency $\omega_0 = (0.7463 \pm 0.0223)Hz$. Using the procedure outlined in §ii.1.6, the damping time $\tau_{avg} = (1.440 \pm 0.0036)s$. Furthermore, by plotting Lissajous figures, the resonant frequency of a damped and driven harmonic oscillator was solve $\omega_R = (0.7160 \pm 0.0005)$ Hz. With these two values, the quality factor Q can also be solved using one method. Another method is done for comparison, which needed a resonance curve to be plotted. Unfortunately, the results (0.5200 ± 0.0014) and (2.7224 ± 0.0055) did not match; the difference between the two values most likely due to approximations made in the derivations such as the small angle approximations and in the inaccurate choices for the frequencies chosen for the resonance width. Smaller sample rates may aid in increasing the accuracy of choice for the frequency calculations.

For the damped, driven vibrating string, two methods were used to compare the values of the wave speed across a stretched length of vibrating string. One method used the tension applied on the string via a pulley and the linear mass density of the vibrating string. Another method used the length a pulse traveled in a reflection and the time interval resulting from sharp rises in the amplitude of the vibrating string. For both methods, the values, corresponding to specific hanging mass, fell within the margins-of-error. Applying the heaviest hanging mass $(0.4000 \pm 0.0005)kg$ on the pulley, Lissajous figures were plotted to determine the fundamental resonant frequency $f_0 = (7.4710 \pm 0.0005)Hz$. In order to study higher modes, approximations based on multiples of the fundamental resonant frequency were used to drive the vibrating string. Additionally, the frequencies for the modes were predicted and compared to the values used in the measurements. Unfortunately, the values did not match within the margins-of-error. However, the difference between the values only deviated by less than 4.5%. Regardless, increasing the precision of digits for the driving frequency can reduced this deviation. Finally, the effects of changing the boundary conditions were studied by adding a node in the middle of the vibrating string. As expected, besides the sudden fluctuation in mode 4, only mode 5 experienced a sudden decrease in amplitude by 2.23% as shown in figure 1.11.

Bibliography

1. Campbell, W. C. et al. Physics 4AL: Mechanics Lab Manual (ver. June 27, 2018). (Univ. California Los Angeles, Los Angeles, California).